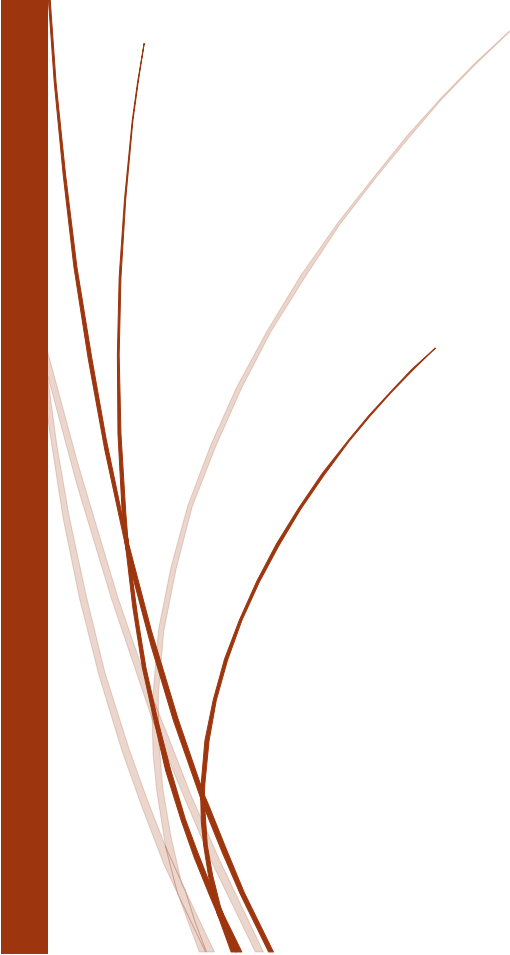




Assignment-5



Created By
~Pintu mer

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

Ping Test -

1. Connect the laptop and phone to the same Wi-Fi network.
2. On the phone, find its IP address from Wi-Fi settings.
Example: 192.168.1.5
3. On the laptop, open Command Prompt.
4. Type:

`ping 192.168.1.5`

5. Press Enter and wait for the result.
6. If you see Reply from..., both devices can communicate.
7. If you see Request timed out, communication is blocked or the devices are not reachable.

Result: The ping test successfully checks communication between the laptop and phone on the same Wi-Fi network.

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Traceroute Test - Simple Steps

1. Connect your device to the Internet.
2. On Windows, open Command Prompt.
3. Type:

`tracert www.youtube.com`

4. Press Enter and wait for the result.

```
Tracing route to www.youtube.com [2001:4860:4829:400::]
over a maximum of 30 hops:

  1      3 ms      3 ms      3 ms  2402:3a80:1eba:9cf7::9e
  2     34 ms     24 ms     26 ms  2402:3a80:1eba:9cf7:0:b:5a94:c040
  3     58 ms     27 ms     37 ms  fd00:abcd:abcd:113::1
  4     90 ms     28 ms     47 ms  fd00:abcd:abcd:203::1
  5     26 ms     26 ms     26 ms  fd00:abcd:abcd:203::2
  6     99 ms     26 ms     36 ms  2400:5200:1000:35::18
  7    102 ms     38 ms     38 ms  2400:5200:401:a::1f
  8    204 ms     31 ms     36 ms  2001:4860:1:1::fe8
  9    218 ms     35 ms     41 ms  2001:4860:4829:400::

Trace complete.
```

Result:

➔ Traceroute has 9 Hops to Travels.

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Ping Test After Disconnecting -

1. Disconnect the phone/device from the Wi-Fi.
2. On the laptop, open Command Prompt.
3. Type:
`ping 192.168.1.5`
4. Press Enter.
5. You will usually see:
"Request timed out."
6. This means the device is not connected or cannot be reached.

Result: Ping failed because the device was disconnected from Wi-Fi.

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

IP Address Table - Simple Steps

1. Connect your laptop to home Wi-Fi.
2. Open Command Prompt.

3. Type:

```
arp -a
```

4. Press Enter.

5. Note the IP addresses shown.

6. Identify each IP address by checking which device it belongs to.

Simple Table

IP Address	Device	Person
192.168.1.1	Wi-Fi Router	Home
192.168.1.2	Laptop	Me
192.168.1.3	Smartphone	Friend
192.168.1.4	Smart TV	Family
192.168.1.5	Smartphone	Brother

Result:

- ❖ The arp -a command displays the IP and MAC addresses of devices that your laptop has recently communicated with on the local network.

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Hint: Include screenshots or command outputs if possible.

❖ Network Setup and Connectivity Testing Report

1. Network Setup

First, I connected my laptop and mobile phone to the same Wi-Fi network. I checked that both devices were connected to the same network. I also checked the IP address of the mobile phone from the Wi-Fi settings.

2. Ping Test

I opened **Command Prompt** on my Windows laptop. I used the following command:

```
ping 192.168.1.5
```

Here, 192.168.1.5 was the IP address of the mobile phone.

The result showed “**Reply from...**”, which means the laptop and mobile phone were able to communicate with each other.

3. Traceroute Test

Next, I tested the path to YouTube using the `tracert` command:

```
tracert www.youtube.com
```

The command displayed several network hops between my laptop and YouTube. I noted the number of hops shown in the output.

4. Disconnected Device Test

After that, I disconnected the mobile phone from Wi-Fi and ran the ping command again:

```
ping 192.168.1.5
```

This time, I received “**Request timed out.**” This showed that the laptop could not communicate with the disconnected phone.

5. ARP Test

Finally, I opened Command Prompt and used:

```
arp -a
```

This command displayed the IP and MAC addresses of devices that my laptop had recently communicated with on the local network.

6. Result and Conclusion

The network setup was successful. The **ping test** confirmed communication between devices on the same Wi-Fi network, while **tracert** showed the path from my laptop to YouTube. When the phone was disconnected, ping failed with a “**Request timed out**” message. Overall, these tests helped me understand how devices communicate on a network and how to check network connectivity.

Screenshot/Command Output:

Screenshots of the `ping`, `tracert`, and `arp -a` commands can be attached below the respective sections.